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Research report

Protean behavior under barn-owl attack: voles alternate between freezing and fleeing and spiny mice flee in alternating patterns

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Abstract

When attacking a spiny mouse in an experimental arena, a barn owl launched a few attacks from distant perches, made repetitive short-distance swoops in each attack and remained in the vicinity of the prey while chasing it. The spiny mouse fled in response, and typically oriented to face the owl whenever it stopped. When attacking a vole, the barn owl performed a greater number of attacks from distant perches, and left the vicinity of the prey after a few short-distance chases or capture attempts. Voles responded to these attacks in unspecific combinations of freezing and fleeing, and did not turn to face the owl when they stopped. Four conclusions are drawn from these encounters. First, two strategies characterized these predator–prey interactions; in one, both predator and prey continuously maintained awareness of each other's location; whereas in the other they continuously attempted to avoid the attention of the other. Second, responses of spiny mice and voles were a manifestation of protean behavior, with spiny mice fleeing in an alternating pattern and voles alternating between running and freezing. Third, locomotor response to owl attack comprised behavior that is an augmentation of normal behavior, with voles clinging to the walls and spiny mice running with frequent and irregular changes in direction. Fourth, the different defensive responses accord with the motor capacities and habitat of each rodent species. All in all, these results demonstrate the dynamic and multidimensional nature of predator–prey interactions.

Keywords: Predation risk; Defensive behavior; Risk assessment; Aggressive behavior; Orientation; Freeze and flee; Spiny mice; Voles

1. Introduction

In the face of life threat, the immediate behavioral responses of a prey animal may include: (i) *freezing*—remaining immobile, typically while crouching and sometimes also relying on camouflage, in order to evade the attention of the predator [6,20,21,35,42]; (ii) *fleeing*—galloping away from the vicinity of the predator [5,8]; and (iii) *fighting* (or *defensive threat*)—heading toward the predator in agonistic display. Defensive fighting occurs only when the prey cannot avoid the predator [30]. Freezing and fleeing were previously described in a variety of prey species, from hermit crabs [36] to deer [40]. For example, a white-tailed deer (*Odocoileus virginianus*) typically flees, flagging its tail [40]; whereas a jerboa (*Jaculus jaculus*, a small bipedal rodent) typically crouches motionless, hiding its white ventral fur and exposing its yellowish dorsal fur to match the desert sand [21]. Response in different individuals of the same species under similar conditions may also dichotomize to freezing or fleeing, as seen in voles exposed to a silhouette of hawk [13,16]. This dichotomy was also observed in the same individual animals under different circumstances: woodmice (*Apodemous mystacinus*) either freeze or leap when exposed to stoats (*Mustela ermina*), [14] but scamper away when exposed to other predators [5,25]. Freezing or fleeing may also vary with age: young white-tailed deer tend to freeze when exposed to a predatory risk, whereas adults typically flee [40]. This conservation of vital functions across species probably reflects a homology in defense systems and their controlling mechanisms.

The neural systems underlying defensive behaviors have been extensively investigated, revealing anatomical differentiation among these systems at both forebrain and midbrain levels (see Neurosci. Biobehav. Rev. 21 (6) for several reviews on various perspectives of defensive behavior). These findings suggest that different defensive behaviors (e.g., freezing or fleeing) may result from independent biobehavioral systems, albeit having a common evolutionary focus on defense against danger. Consequently, various

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defenses may be selectively responsive to drugs acting on particular neurotransmitter systems or subsystems [3,41]. Thus, understanding the fine-grained structure of defensive behavior may provide insights into controlling mechanisms and neural systems that underlie threat-related behaviors.

Three hypotheses were posed to explain the question of what makes an animal ‘freeze’ or ‘flee’: (i) *individual differences* that may also be echoed in other behavioral perspectives such as baseline activity and social rank [13]; (ii) *distance to the predator*, with distant predator inducing freezing, a closer predator inducing fleeing, and an even closer predator inducing defensive fight [18,34]; and (iii) *access to shelter*, promoting flight to a safer area, or freezing when no shelter is available [4]. Accordingly, it has been suggested that discrete and localizable threat source (i.e., visible predator) promotes flight and defensive threat/attack, while amorphous/difficult to locate threat source (i.e., owl call) promotes freezing and risk assessment [2].

The above studies have typically described a single defensive response, typically classifying the overall response as either freezing or fleeing, but not both. A transition from freeze to flee was not described as spontaneous, but dependent on other factors such as approach of the predator. Certainly, it is impossible to freeze and flee simultaneously, and at any given moment a prey can execute only one of these defenses [29]. However, here we assumed that only freezing or only fleeing would make the prey predictable and vulnerable. Rather, in order to elude the predator successfully, the prey should also take advantage of its motor capacities and habitat structure, performing unpredictable defensive responses intended to confuse the predator. Consequently we scrutinized moment-by-moment real encounters between a barn owl (*Tyto alba*) and two rodent species: the social (Gunther’s) vole (*Microtus socialis*) and the common spiny mouse (*Acomys cahirinus*). The rationale for selecting these rodent species was that they represent extremes in the spectrum of defensive responses. Voles have been shown to display a conspicuous response to owl calls, by either freezing or fleeing, with concomitant increase in their corticosteroid levels as a physiological indicator of stress [13]. In contrast, spiny mice did not alter their activity when exposed to owl calls [21], despite increasing their corticosteroid levels [13]. Encounters between a barn owl and vole or spiny mouse were analyzed in order to provide a dynamic and detailed description of defensive behavior, to refine the simplistic dichotomy of freeze or flee, and to provide a heuristic explanation of the advantage of each type of defensive response.

2. Materials and methods

2.1. Study animals

The social (Guenther’s) vole (*Microtus socialis guentheri*) weighs 37–50 g and is 11 cm long, plus a 2-cm tail. It is a burrow-dwelling rodent that feeds on seeds and green vegetation. The common spiny mouse (*Acomys cahirinus*) weighs 38–44 g and is 11 cm long, plus a 10-cm tail. It is an agile rodent, common on rocky mountains, where it shelters in rock crevices [37,39]. In this study we used videotapes that had documented encounters between voles and spiny mice in another study [9]. Briefly, the videotapes comprised individual encounters of a barn owl with 13 voles and 13 spiny mice; all these encounters were re-analyzed in the present study. The rodents were bred in captivity in colonies at the research zoo of Tel-Aviv University and had been previously used in other studies [11,12]. Several weeks before testing, the rodents of each species were housed in groups of 5–10, in metal cages measuring 40 cm × 70 cm × 25 cm, located outdoors in the zoo yard under natural (uncontrolled) temperature and light conditions. Overturned ceramic pots and wooden boxes were placed in each cage to provide shelter. Seeds and diced fresh vegetables were provided ad lib. Spiny mice were also provided with live fly larvae. Based on years of experience in maintaining colonies of voles in our zoo, provision of water is unnecessary when sufficient fresh vegetables are provided.

The barn owl (*Tyto alba*) weighs 250–315 g and is 31.5–36 cm long with 28 cm span of elliptic wing. Dorsal feathers are golden-yellow and the ventral feathers are white with scattered brown dots. Barn owls are efficient raptors that feed mainly on rodents. The initial detection of prey location relies on hearing the sounds generated by prey movement, and is followed by visually pinpointing the prey with sharp night-vision. Barn owls then swoop down on the prey from a perch or on the wing, catching it with their spiked talons and killing it in seconds. A colony of barn owls is kept in the research zoo of Tel-Aviv University, and is fed with freshly killed (and from time to time also live) chicks and mice, obtained from surplus stock from the animal quarters and from chicken-incubators. Thus, these captive barn owls are accustomed to preying on live rodents. Because of the high fecundity of voles and spiny mice (early maturation, short weaning period, frequent all-year-round breeding in captivity) and the extensive use of these species in other research projects, surplus of these rodents is also used to feed the owls. One adult male barn owl was selected as the predator in the present experiment. We chose this specific barn owl since preliminary observations on our captive barn owls revealed that the selected individual had relatively short latency in attacking live prey. Throughout the experiment period the owl was provided with one rodent per night, which is the regular feeding schedule at the zoo.

2.1.1. Ethical consideration

This study was carried out under the regulations and approval of the institutional committee for animal experimentation (permit #L-02-40). The design carefully followed the *guidelines for the treatment of animals in behavioural research and teaching* [45] with special emphasis on the guidelines for staged encounters as outlined by [24], as follows.

Sample size was minimized to six to seven animals per pattern of response (freeze or flee). Videotaping at peak activity of the owl, when latency to attack was short (averaging 7 min) minimized the duration of predatory encounters. The rodents were relatively relaxed throughout the greater part of the test, with behavior that resembled that of controls that were not exposed to the owl [9]. Furthermore, even when a refuge was provided and the presence of an owl was apparent, the rodents continued to locomote outside the shelter. Owl predation is fast, averaging 1.5 s from onset of attack to capturing and killing the rodent. The rodents were "recycled" from previous studies: they had been first studied as pups in studies on parental behavior [27], and in studying locomotor behavior [11,12], and only then allocated to the study of predation. Owls at the research zoo are fed on rodents, including the species used in this study. Therefore, observations were basically a matter of setting up equipment to videotape the owl during feeding, without trapping additional wild animals or sacrificing more than the regular dietary requirements of the owls and of other experiments.

2.2. Apparatus

Observations took place in a 5 m × 5 m × 4 m aviary with perches and a small tree, where a barn owl was housed for several weeks before testing. A 1 m × 1 m enclosure with 60-cm high walls and a transparent glass front was placed in the center of the aviary, used as the rodent's enclosure. Two infrared-sensitive video cameras (*Sony* TRV23E; *Ikegami* ICD47E) that allow vivid shots in complete dark were installed in the aviary, one encompassing the entire cage, providing information on the location of the owl, and the other focusing on the enclosure, providing a close view of the rodent. Two IR lights (*Tracksys*, IR LED Illuminator, UK) that emit light in a range invisible to owls and rodents followed the directions of the cameras. A video mixer (*Panasonic* WJ-AVE5) was fed with the video signals of both cameras, providing a single composite picture, showing the attacking owl and the defending rodent in the same frame (=same time), thus enabling us to monitor the behavior of the owl and the immediate response of the rodent from moment to moment. A time-code generator (*Telcom Research*, Canada; T-5010 + T800) that marks every video frame (25 frames/s) was added to the composed video signal, which was stored in VHS format VCR (*LG* C20W).

2.3. Procedure

Tests (one per night) took place between 11.00 p.m. and 2.00 a.m., since during these hours the latency for owl attack is relatively short. At the beginning of a test period, an experimenter switched on the equipment, released a rodent into the enclosure, and left the aviary area. The next morning, the videocassette (180 min) was collected and scanned for the owl–rodent encounter, which was then analyzed.

2.4. Behavioral analysis

A moment-by-moment analysis was carried out, starting with initiation of the first attack of the owl and ending with the successful capture of the rodent. In this analysis, six parameters were scored:

1. *Attacks* (=number of times the owl entered the rodent's enclosure): An attack started when the owl entered the enclosure and ended when it left. Duration and number of attacks were measured.
2. *Capture attempts*: While inside the enclosure, the owl could make repetitive attempts to catch the rodent (e.g., by approaching, chasing, or swooping). The incidence of these attempts was scored, together with the response of the rodent to each one.
3. *Response of the rodent to capture attempt*: For each capture attempt, the response of the rodent was scored as '*freeze*' or '*flee*.' When the rodent was immobile for at least 5 s it was considered as freezing. When the rodent made a fast move it was considered as fleeing.
4. *Time between successive attacks*: Time elapsed between the end of one attack (when the owl left the rodent enclosure) and the beginning of the next (when the owl re-entered the enclosure).
5. *Catch duration*: Time elapsed between the beginning of the first attack and successful capture of the rodent.
6. *Oppositions*: These were scored only when the owl was inside the rodent enclosure, the opponents were close to one another, and both were momentarily immobile. Oppositions have been found useful in studying interaction between two animals [15,19,43] in describing which body part is closest to the opponent. For this description, five oppositions were defined for both the owl and the rodent: front, front-side, side, side-back, and back. When, for example, the rodent and the owl faced each other, 'front' was scored for both. When the rear side of the rodent was facing the head of the owl, the score was 'back' for the rodent and 'front' for the owl.

2.5. Statistics

As noted above we used only a single specific barn owl in order to establish consistency in threat. Therefore, the presented statistical tests were aimed at defensive behavior in voles and spiny mice, not meant to indicate the level of confidence regarding inferences to owls in general, but rather to convey some measure of consistency in the performance of the owl during tests. In Table 1, two-way analysis of variance was performed on $\ln(x + 1)$ of the raw data that was extracted from the encounters (x = each scored datum, addition of 1 was to reduce the effect of zeros on statistical tests). Within group factor was rodents caught in first attack versus rodents caught in repetitive attacks. Between group factor was species (voles versus spiny mice). Since these data are strictly linked, a Bonferroni correction was applied, setting

Table 1
Voles and spiny mice that were caught in the first attack compared with those that were caught only after repetitive attacks

	Caught in first attack		Caught in repeated attacks	
	Spiny mice	Voles	Spiny mice	Voles
Attacks	1 ± 0	1 ± 0	2.4 ± 0.27	4.4 ± 1.15
Capture attempts	6.75 ± 1.94	4 ± 2.24	13.6 ± 4.59	13.4 ± 2.25
Duration to catch (s)	11.5 ± 3.63	11.19 ± 9.21	250 ± 59	880 ± 523
Freezing responses	0.13 ± 0.13	0.75 ± 0.17	0 ± 0	4.4 ± 1.25
Fleeing responses	6.63 ± 2	3.25 ± 2.35	13.6 ± 4.59	9 ± 1.97

Values represent mean ± S.E.M. In three parameters there was a significant difference between rodents caught in first attack compared with those caught in repetitive attacks: number of attacks (trivial since attacks were ranked according to this category; $F_1 = 52.9$, $P < 0.0001$), duration to be caught ($F_1 = 84.4$, $P < 0.0001$) and number of freezing responses ($F_1 = 25.5$, $P < 0.001$). For freezing response, there was also a significant between-group (species) effect ($F_1 = 42$, $P = 0.0002$), and a significant species × number of attacks interaction ($F_{1,1} = 25$, $P < 0.001$). There was no significant difference, neither between species nor between first and repetitive attacks in the number of capture attempts and in the number of fleeing responses.

alpha level to 0.01 (0.05 divided by the five parameters that were compared). Oppositions in the owl and the two species were compared in a two-way ANOVA (within-group factor: direction of opposition; between group factor: species), followed by Tukey HSD test. Alpha level was 0.05.

3. Results

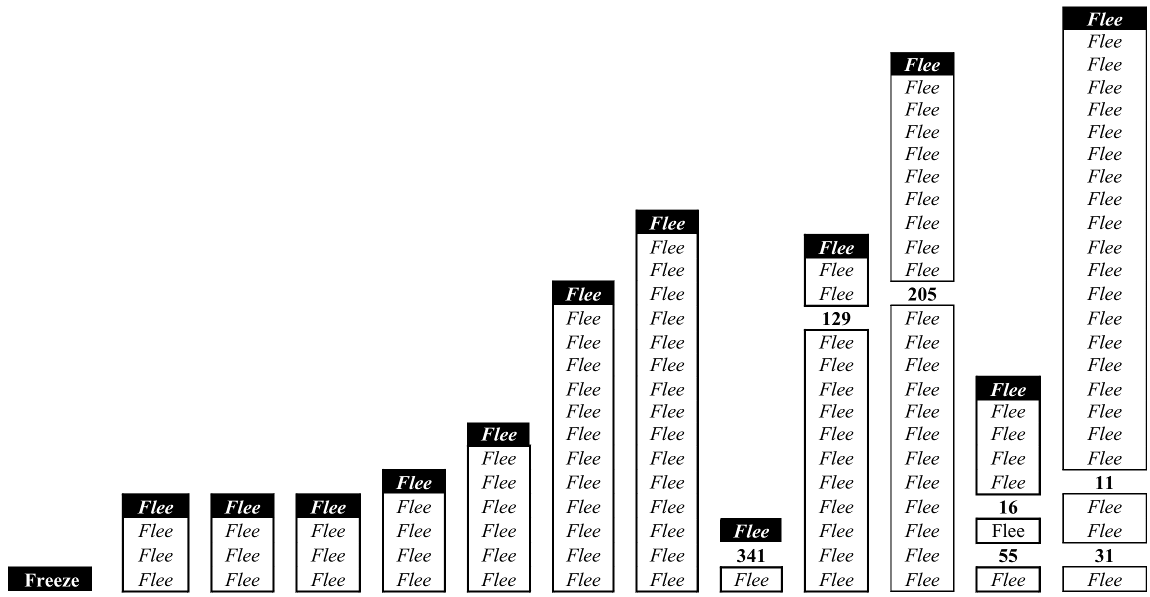
3.1. Temporal structure of owl–rodent encounter

Fig. 1 depicts the encounters between the owl and spiny mice. Each column represents the encounter of the owl with one spiny mouse (1–13), ranked from left to right according to the number of attacks (blocks) in each encounter (i.e., number of times the owl entered the rodent's enclosure). Numerals between two adjacent blocks show time between successive attacks. Each attack includes at least one capture attempt. Capture attempts executed during the same attack (owl remaining in rodent enclosure) are shown in a frame (block) from bottom to top, followed by the time between end of one attack and beginning of subsequent one. For example, in spiny mouse # 13 (right-hand column), the owl made a first attack with one capture attempt during which the spiny mouse responded by fleeing. The owl then left the rodent enclosure for 31 s, and then initiated a second attack in which it made two capture attempts, with the spiny mouse responding to both by fleeing. The owl again left the rodent enclosure for 11 s, and then made a third attack (third block) in which it performed 20 capture attempts, culminating in a successful catch (depicted on a black background). Thus, spiny mouse # 13 was caught after three attacks (represented by three blocks) during which the owl made 23 capture attempts (the content of the three blocks), to which the spiny mouse consistently responded by fleeing. These details are summarized at the bottom of column 13.

Comparing the 13 spiny mice (columns) reveals that the response to all capture attempts was to flee, except for one incidence in animal # 1. The owl typically executed several capture attempts within each attack, and there was a large variation in the number of attacks and capture attempts executed by the owl. Eight spiny mice (columns 1–8) were caught in first attack. One of these (left-hand column-13) was caught in first capture attempt (during first attack). The other seven animals were caught in repeated capture attempts that comprised the first attack. In other words, spiny mice 1–8 were caught in the first time that the owl entered their enclosure, but typically only after several capture attempts. On five spiny mice (columns 9–13), there were 2–3 attacks, meaning that the owl left the enclosure and then launched another attack(s) from far roosts. Elapsed time between attacks varied from 10 to 350 s. Thus, overall pattern was of a few attacks, each with several capture attempts. The number of attacks and capture attempts did not necessarily reflect more (or less) time taken for successful capture. It could take more time to catch one spiny mouse in two attacks each with two capture attempts (column 9), than to catch another one in three attacks comprising 23 capture attempts (column 13). The testing order depicted at the bottom of the figure indicates that there was no effect of testing order on the duration to successful catch or the behavioral patterns.

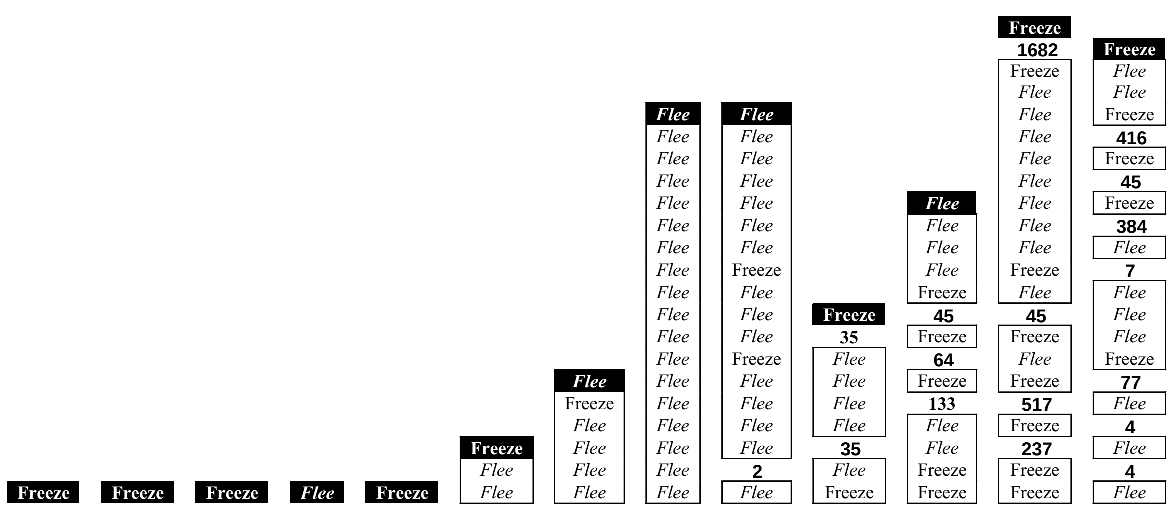
Examination of the data from encounters between owl and voles (Fig. 2) revealed a different pattern, with large variation in the number of attacks and capture attempts executed by the owl. Eight voles were caught by the owl in the first attack (columns 1–8), with five of them being caught in the first capture attempt (columns 1–5). Five voles were caught only after 2–8 repeated attacks (columns 9–13), each comprising 1–16 capture attempts. Time elapsed between successive attacks ranged from 2 to 1700 s, and time until a vole was caught increased with the number of attacks: capture was achieved within several tenths of seconds in a single attack, within several hundred sec in 2–4 attacks, and within several thousand sec in 5–8 attacks.

The response of voles to capture attempts was inconsistent, with some individuals performing freezing, fleeing, or both (e.g., column 12). Overall, there were 28 freeze responses and 71 flee responses. However, there was no difference in the duration to capture voles that froze in response to the first capture attempt (bottom of columns 1, 2, 3, 5, 10, 11, and 12), and voles that fled in response to the first capture attempt (bottom of columns 4, 6, 7, 8, 9, and 13).



Spiny mouse #	1	2	3	4	5	6	7	8	9	10	11	12	13
Attacks	1	1	1	1	1	1	1	1	2	2	2	3	3
Catch attempts	1	4	4	4	5	7	13	16	2	14	22	7	23
Duration to catch	1	5	6	6	9	13	27	25	371	332	290	93	162
Freezings	1	0	0	0	0	0	0	0	0	0	0	0	0
Fleeings	0	4	4	4	5	7	13	16	2	14	22	7	23
Testing order	12	11	10	9	13	7	4	3	5	6	2	8	1

Fig. 1. Interactions between 13 spiny mice and a barn owl. Each column represents the responses of one spiny mouse to capture attempt by the owl arranged from bottom to top and depicted as 'freeze' or 'flee.' Capture attempts that were executed in the same attack (i.e., while the owl was in the spiny mouse's vicinity and had not left the rodent enclosure) are enclosed in frames (blocks). Time elapsed between successive attempts is depicted (in seconds) between attacks (blocks). Data at the bottom of each column summarize the interactions.



Vole #	1	2	3	4	5	6	7	8	9	10	11	12	13
Attacks	1	1	1	1	1	1	1	1	2	3	4	5	8
Catch attempts	1	1	1	1	1	3	6	18	17	7	11	18	14
Duration to catch	0.5	1	1	2	2	2	10	71	78	127	489	2602	1104
Freezings	1	1	1	0	1	1	1	0	2	2	5	8	5
Fleeings	0	0	0	1	0	2	5	18	15	5	6	10	9
Testing order	8	12	11	9	2	1	4	6	3	13	5	10	7

Fig. 2. Interactions between 13 voles and a barn owl. Each column represents the sequence of one vole, with the response of that animal to capture attempt of the owl arranged from bottom to top and depicted as 'freeze' or 'flee.' Capture attempts that were executed in the same attack (i.e., while the owl was in the vole's vicinity and had not left the rodent enclosure) are enclosed in frames (blocks). Time elapsed between successive attempts is depicted (in seconds) between attacks (blocks). Data at the bottom of each column summarize the interactions.

Voles that responded with both freezing and fleeing (10, 11, and 12) survived longer than voles that mainly fled (4, 6, 7, and 8) or mainly froze (1, 2, 3, and 5).

The behavior of the owl in attacking voles as opposed to attacking spiny mice (bottom sections of Figs. 1 and 2) was also different. On voles it carried out more attacks, each with fewer capture attempts, whereas on spiny mice it launched fewer attacks with more capture attempts. Pauses between successive attacks were longer in voles compared with spiny mice. Altogether, spiny mice were caught faster than voles (15.7 ± 2.2 min and 30.6 ± 5.2 min, respectively, from the moment the owl left its perch until it caught the rodent).

The findings also reveal a substantial difference between the animals caught in the first attack (rodents 1–8 in Figs. 1 and 2) and those caught after repeated attacks (rodents 9–13 in Figs. 1 and 2). Data on these two subgroups were averaged and are given in Table 1. As shown for the animals caught in the first attack, the duration to capture was similar for spiny mice and voles, with a higher number of capture attempts and fleeing responses in spiny mice. In the animals caught during the following repeated attacks, the number of capture attempts was similar, but these were divided over more attacks in voles, resulting in longer duration to capture voles compared with spiny mice. Fleeing was prevalent in both rodent species, whereas freezing was observed mainly in voles.

3.2. Oppositions between owl and rodent

The above results revealed the temporal (sequential) structure of owl attack and the corresponding defensive response of the rodents. Further analysis showed that encounters also comprised non-locomotory periods during which the rodent and the owl were immobile, until movement by one reinstated the chase. To uncover possible differences at this stage, the opposition between the owl and the rodent was scored whenever a rodent ceased to locomote while the owl was inside the rodent's enclosure. Fifty-one owl/vole oppositions and 110 owl/spiny mouse oppositions were recorded, indicating for both owl and rodent the side of their body closest to the opponent (Fig. 3a). The relative incidence of these oppositions is shown in Fig. 3b, revealing a difference between oppositions (two-way ANOVA; $F_{4,68} = 4.92$, $P = 0.001$) and interaction of the species (vole, spiny mouse) and oppositions ($F_{4,68} = 8.25$, $P < 0.001$). Tukey Honest Significant Tests showed that the owl adopted a frontal opposition in encountering either voles or spiny mice. In other words, the owl tended to turn and face its prey. Spiny mice also adopted frontal opposition, tending to turn toward the owl whenever they stopped. Voles, however, showed no preference for a specific opposition, reflecting their tendency to freeze suddenly regardless of their opposition toward the owl. As suggested in the following discussion, this difference in oppositions may account for the faster capture of spiny mice under the present test conditions.

4. Discussion

The present study has shown that during encounters with spiny mice, barn owl launched only a few (1–3) attacks, from distant perches. In each attack, the owl typically

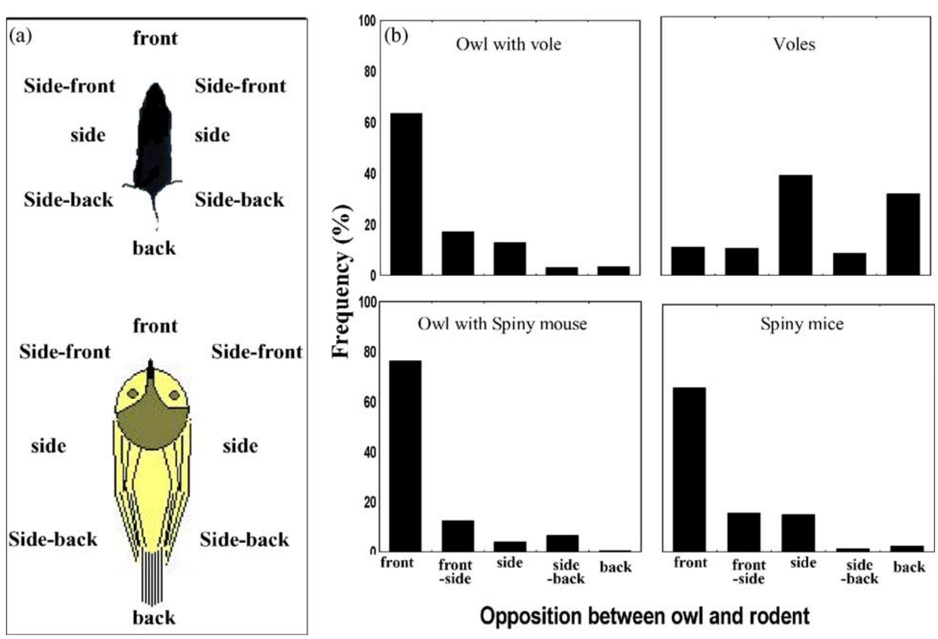


Fig. 3. The oppositions (a) of the rodent (top) and the owl (bottom) were scored by indicating the side of the body closest to the opponent. As shown, five oppositions were defined for each opponent (a). The relative incidence of oppositions (b), is shown for owl (top left) and voles (top right), and for owl (bottom left) and spiny mice (bottom right).

remained in the vicinity of the spiny mouse while chasing it and making repetitive short-distance capture attempts. Spiny mice fled in response, and typically oriented to face the owl whenever they stop. In encountering voles, barn owl made more (1–8) attacks from distant perches. In each attack on voles the owl typically performed only a few (if any) repetitive short-distance capture attempts and then returned to a distant perch. Voles responded by freezing, fleeing, or both, but did not turn to face the owl when they stopped. We suggest that these behavioral patterns throw new light on the concept of freezing and fleeing as defensive responses, being part of a more general strategy of frequent and random changes that may confuse the predator. We also suggest that each defensive response accords with the motor capacities and habitat of each rodent species.

Freezing and fleeing are generally considered as mutually exclusive defenses (see Section 1). However, following the moment-by-moment response of voles illustrated that exclusiveness of these responses is only momentary, and the same vole could abruptly switch from freezing to fleeing and vice versa. Spiny mice, in contrast, mainly fled and rarely froze (only 1 out of 121 responses in spiny mice was freezing). Summing these momentary responses into an inclusive definition of the individual response as freezing or fleeing, as applied in previous studies [13], is therefore oversimplistic and does not convey the complete nature of the response and possible advantages of using these defensive patterns. We suggest redefining the classification of defensive responses according to two main patterns: (i) employing both freeze and flee, as do voles; and (ii) fleeing from a predator, as do spiny mice.

Previous study revealed that owls could more easily catch a stationary rodent, compared with a moving one that imposes maneuvering and chasing [38]. Fleeing therefore seems advantageous if a prey has the capacity to escape and can get to a shelter before the owl re-attacks. In the present study, spiny mice did not have access to shelter, and by their incessant locomotion stimulated owl attacks, resulting in being caught faster than voles. However, in the wild, fleeing is appropriate for the nimble spiny mice [10,33] that live and forage in crevices and spaces between and under boulders, spending only little time in the open [26]. In other words, agility, fast running, and nearby shelters make fleeing the more appropriate response for spiny mice during owl attack. Indeed, spiny mice progress in brief short leaps, covering two-fold longer distances compared with voles [12,13]. Accordingly, when spiny mice were exposed to predatory threat in previous studies, they either increased activity and fled [1] or preserved their ongoing high level of activity [13,21].

Voles are clumsier than spiny mice, with walk and trot gaits [10] and slow locomotion [44]. In addition, they live in relatively open spaces [31] where, once outside their burrows, they are less likely to escape an attacking owl. Thus, voles may rather freeze, eliminating the auditory and visual cues that owls use in pinpointing prey [32], and rely on the camouflage of their brown fur that matches the color of the heavy soil they inhabit. In other words, if a vole freezes before being spotted, the owl may not be able to locate it. However, if seen by the owl before freezing, the vole may become a sitting target. This duality probably led voles to use both freeze and flee defenses under owl attack. Moreover, upon noticing the owl, a vole might not know whether or not it has been seen. In that case, alternating between freezing and fleeing may be an optimal response, combining disappearance by freezing together with not being a stationary target if freezing fails. Chances of escape further increase by the variability in the defensive response of different voles: only freezing (voles 1, 2, 3, and 5 in Fig. 2), only fleeing (voles 4 and 8 in Fig. 2), switching several times between freezing and fleeing (voles 6, 7, 9, and 10), and alternating frequently between these responses (voles 11, 12, and 13 in Fig. 2). Thus, voles respond in a spectrum of combinations of freezing and fleeing, and their response should neither be considered as just freeze or flee [13,21], nor be defined according to the initial response [9].

A similar dependence of defensive response on motor capacities and habitat structure was previously described in deer species [30]. White-tailed deer (*Odocoileus virginianus*), which inhabit forests and are fast runners, tend to flee when encountering coyotes (*Canis latrans*), whereas mule deer (*Odocoileus hemionus*), which live in relatively open spaces and are moderate runners, tend to freeze or flee. Thus, as in spiny mice and voles, better motor capacities and habitat that provides nearby shelter encourage a flight response, while limited speed and open habitat conduce to freeze and/or flee [9].

The suggested patterns of fleeing or of combining freezing and fleeing are further illustrated in the oppositions between the owl and the rodents, as described in the results. Voles stopped and froze suddenly, as if pausing mid-activity and remained immobile for a while. In contrast, when spiny mice stopped, they turned to face the owl. Fig. 4 illustrates how these behaviors may influence the success of the owl in catching them. As shown, the owl rotates its head and/or trunk to follow an escaping vole. When the vole suddenly freezes, the owl initially continues to track the imaginary trajectory of its movement, until noticing that it is no longer following the vole and returning to the last spot where the vole was seen. At this point, the vole is immobile, generating no sound or visual cues that may expose it, and vanishing in the darkness with its dark fur against the background of the brown-red soil it inhabits. Losing the vole, the owl now flies back to a perch until the vole resumes locomotion and stimulates another attack. Accordingly, the structure of barn owl attack on voles was launching attacks from far roosts when the voles moved, but making only few capture attempts or chases when the owl was in the vicinity of freezing voles.

When encountering a fleeing spiny mouse, the owl rotates its head and/or trunk toward it as with the vole. When the prey stops, the owl continues to track the imaginary trajectory of its movement until noticing that it is not following

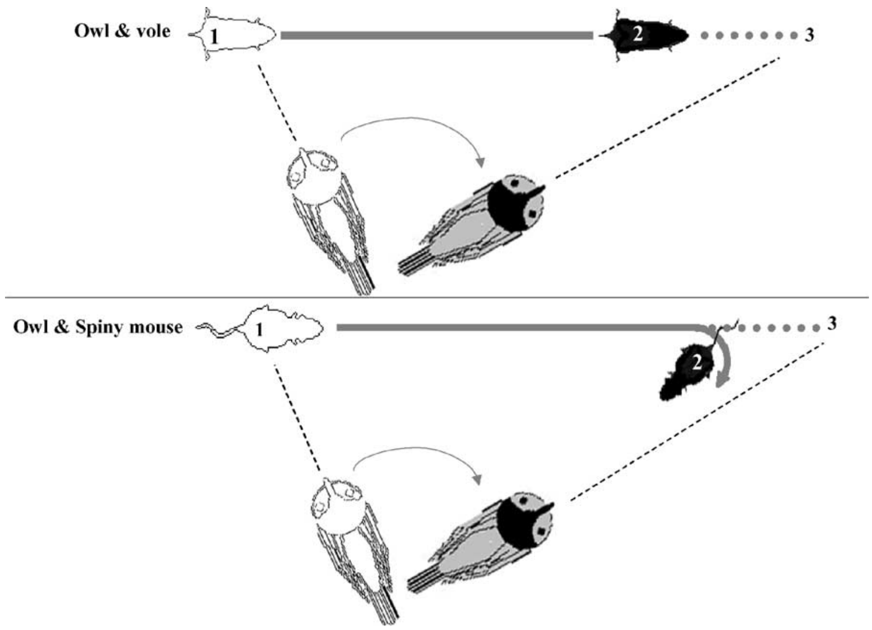


Fig. 4. A model of how an owl tracks a vole (top) and a spiny mouse (bottom) during a close-distance encounter. The movement of orienting toward the owl in the spiny mouse exposes it to the owl, whereas the immobility of the vole facilitates eluding the owl's attention.

it anymore, and returning to the last spot where the spiny mouse was seen. At this point, the prey is now turning to orient toward the owl, thereby exposing its location. The owl therefore remains in the vicinity of the spiny mouse, initiating repetitive capture attempts that stimulate another flee response by the spiny mouse. The result is only a few attacks on a spiny mouse with each attack comprising repetitive capture attempts, in contrast with several repetitive attacks on a vole with each attack comprising only a few capture attempts. The different defensive response of each rodent species thus elicits a different pattern in owl attack. Conceptually, two opposite strategies (mechanisms) underlie these predator–prey interactions: owl and spiny mouse, by virtue of their constant movement, are continuously aware of the location of the opponent, whereas owl and vole continuously attempt to avoid the attention of the opponent.

So far we have suggested that there are two patterns of defensive behavior: (i) combining freezing and fleeing, and (ii) fleeing. We now suggest that these two defences share the underlying mechanism of *Protean Behavior*, named after the Greek God *Proteus* who frequently changed his appearance and character in order to confuse others. Protean behavior is based on frequent and irregular behavioral changes that confuse the predator, allowing the prey to elude the attack [22,23]. Group animals under attack scatter to escape in various directions, confusing the predator, which has to focus on one prey animal and ignore the others. Protean behavior of an individual prey animal under attack is reflected in either zigzag running [17] or irregular change in behavioral patterns [7]. We suggest that these forms of protean behavior are also applicable to the present results, with switching between freezing and fleeing being a change in behavioral patterns, and fleeing in spiny mice being the zigzag run.

Studies on motor behavior in voles and spiny mice illustrated that normally voles tend to progress along straight trajectories whereas spiny mice tend to frequently change the direction of progression. Fig. 5 depicts the trajectories of locomotion in a vole and a spiny mouse, illustrating the tendency of voles to travel along the walls of the arena and

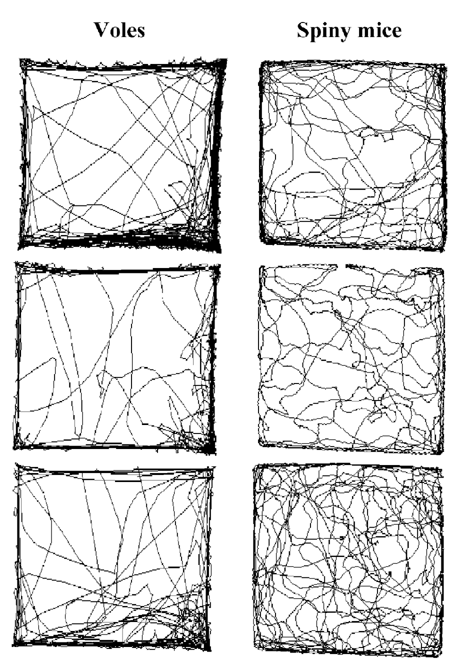


Fig. 5. Trajectories of locomotion of spiny mice (right) and voles (left) during 10 min in a 1 m × 1 m enclosure, as described in previous studies [11,12]. Trajectories are the output of a tracking system (*Ethovision*, NL). Testing took place in an empty and silent room, with the rodents exposed to nothing but the test enclosure.

to cross the center in relatively straight segments that extend from wall to wall [11,12]. The illustration of spiny mice trajectories reveals locomotion both in the center and along the perimeter of the arena, with frequent changes in the direction of progression without straight segments [12]. Thus, defensive locomotion is an augmentation of normal patterns, with voles clinging to the walls and spiny mice moving with frequent and irregular changes in the direction of progression [9]. A mathematical model on the trajectories of flight from a predator suggests that a straight trajectory is advantageous in facing a distant and relatively slow predator, whereas a zigzag trajectory is advantageous when encountering a close or fast predator [17]. In the present study, the same predator (barn owl) attacked slow prey—voles, and fast prey—spiny mice. The zigzag trajectories of the escaping spiny mice are thus in accordance with the mathematical model [17], whereas the straight trajectories of escaping voles are appropriate for a slow prey that relies on a protean combination of freeze and flee responses.

The present results in wild rodents are of fundamental importance in extending the knowledge on defensive behavior, as revealed in laboratory species, to additional species. Defensive response in wild species is vigorous and unequivocal, with the behavior evaluated under circumstances that may commonly occur in nature. These may enable a more precise assessment of defensive responses and factors that control them in laboratory species. Following encounters between barn owl and rodents moment-by-moment, we also show that both the prey and the predator participate in a behavioral interaction that is more complex than those revealed in studies that consider the predator as an abstract source of threat [28]. In the light of these results we suggest that the dichotomy to freezing or fleeing is momentary, and is part of a more general pattern of response in which rodents apply protean behavioral changes that depend on their motor capacity, on the structure of their habitat, and on their normal behavior [9]. Consequently, the manifestation of protean behavior is a dichotomy between fleeing in an alternating pattern in agile species and complex habitats, and combinations of running and freezing in slower species and open habitats.

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